TPCPF: Three-Phase Continuation Power Flow Tool for Voltage Stability Assessment of Distribution Networks With Distributed Energy Resources

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***Abstract*—This work presents a three-phase unbalanced continuation power flow algorithm for voltage stability assessment of distribution systems with high penetration of distributed energy resources (DERs). Analyzing distribution system voltage stability with DER will allow high penetration of renewable energy necessary for the sustainability goals. The developed algorithm can analyze voltage stability for both the meshed and radial systems and the balanced and unbalanced three-phase distribution systems. The developed tool allows the voltage stability analysis to facilitate the planning, operation, control, and distribution system management. The impact of DER on the voltage stability of several test cases has been analyzed considering constant power (PQ) and regulated-voltage (PV) modes of operation for DER units. Moreover, different voltage stability case studies are presented to demonstrate the impact of unbalance, load increment, and network topology on the maximum loading capacity. Results using the IEEE 13-node feeder, the 18-bus balanced shipboard system, the 13-node CIGRE benchmark system, and the 136-bus redial distribution feeder demonstrate that the developed continuation power flow tool can efficiently perform voltage stability analysis for active distribution systems.**

***Index Terms*—Continuation power flow (CPF), distributed energy resources (DERs), distribution stability, voltage stability.**

I. Introduction

VOLTAGE stability analysis is critical for operating the system in a reliable and efficient mode considering the continuous growth in demand driven by environmental, economic, and technical constraints. Voltage collapse has been one of the prime reasons for several significant disturbances in the past [1], [2], and as a result, voltage stability assessment research has received great attention [3]–[5]. Voltage stability is usually considered a local phenomenon, but the impacts could be widespread and could lead to a system-level voltage collapse [6].

Voltage stability of power systems has been a vastly studied topic for transmission system analysis [7], [8]. The continuation power flow (CPF) method has always been considered one of the best methods to evaluate voltage stability [9]–[11]. CPF-based approaches have proven their mettle and have become a necessary tool in many energy control centers. Transmission system operators use CPF to determine the maximum loading capacity and also for detailed contingency analysis. In general, the CPF-based approaches use predictor–corrector steps to trace the P–V curves as the loading multiplier is increased from the base case till the collapse point. As the system becomes stressed, the power flow Jacobian approaches singularity, and CPF uses parameterization and augmentation of power flow Jacobian to ensure convergence under stressed operating conditions.

CPF for distribution networks has received relatively less attention compared to transmission. Most mature transmission analysis tools have well-developed CPF libraries capable of tracing the static voltage behavior. Some researchers have tried extending the conventional forward–backward sweep power flow to develop CPF tools for distribution [12]. The major disadvantage of forward–backward sweep-based CPF is limited applicability in the analysis of meshed distribution networks. In [13], Ju *et al.* have presented a loop-based CPF method for distribution systems, but this algorithm can only be used for radial and weakly meshed networks. In [14], Sheng and Chiang have developed a comprehensive three-phase continuation power flow (TPCPF) tool for distribution networks. The tool developed in [14] uses a modified Newton method for corrector implementation. Dou [15] also consider the traditional Newton method for the development of the CPF tool for distribution networks with high distributed energy resource (DER) penetration. The Newton method in its traditional form has faced convergence issues in the power flow analysis of distribution networks due to their appreciably high R/X ratio. Thus, a CPF tool embedded with a power flow solution method suitable for meshed and radial distribution networks is highly desirable to suit modern active distribution systems' planning and operational analysis needs.

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Traditional power flow methods adopted for transmission system analysis have faced convergence issues when applied to analyze the distribution system. The reason behind unsuitability lies in the special characteristics of distribution networks. The distribution networks possess a comparatively high R/X ratio and are mostly radial in the literature. That, coupled with the multiphase unbalanced operation and increasing penetration of DERs, has resulted in numerical issues when the traditional Newton method is used for distribution networks [16], [17]. The convergence issues are further aggravated by distribution systems moving toward more meshed topologies. This evolving active nature of distribution networks necessitates dedicated power flow algorithms for unbalanced distribution system analysis. The three-phase power flow methods explained in [18] and [19] use the current injection method for power flow analysis in distribution networks. Recent modernization of the electric grid with smart grid revolution has made tremendous modifications in distribution systems' structure. A high degree of unbalance, DER penetration, and multiphase radial feeders make voltage stability a pivotal concern in distribution networks. The three-phase current injection method (TCIM) in [19] is a powerful distribution power flow solver for both meshed and radial networks. This method has also been adopted in commercial-grade distribution solvers such as OPENDSS [20] and GRIDLAB-D [21]. OPENDSS [20] uses a TCIM for analysis of both radial and meshed distribution networks, whereas GRIDLAB-D [20] uses forward–backward sweep for analysis of radial configurations and uses the current injection method for meshed configurations. Because of the superior performance of the current injection method in three-phase distribution analyses, this work considers using the TCIM [18], [19] for the development of the proposed CPF tool for active distribution networks.

This work builds on our previous work [22] and presents the TPCPF tool for voltage stability analysis with the following contributions.

1) Developed formulation and algorithm for CPF for unbalanced three-phase distribution power systems with DERs using the current injection power flow method, the adaptive step predictor–corrector, the PV–PQ switching algorithm, and the nose point convergence criterion.
2) Investigated the effect of DERs in improving the voltage profiles and improving the system loading margin. The impact of unbalance, topology, DER penetration, and DER models is also demonstrated on the system loading margin.
3) Analyzed the performance for both meshed and radial active distribution networks.
4) Validated the performance for four test cases: IEEE 13-node feeder, 18-bus balanced shipboard system, 13-node CIGRE benchmark system, and 136-bus redial distribution feeder.

II. Three-Phase Current Injection Power Flow

The TCIM for power flow presented in [23] is used for the development of the proposed CPF tool. In the TCIM, real and imaginary current injections for each phase are used instead of power injections used in traditional-Newton-method-based power flow. There are a total of $6n$ variables for current injections, where n is the number of buses. The current injection method's Jacobian matrix consists of blocks (6×6) corresponding to PV and PQ buses. Blocks corresponding to PV buses have to be updated at each iteration, whereas the blocks corresponding to PQ buses are constant. At each iteration, only a few block entries have to be updated, enhancing the computation efficiency of the TCIM compared to the Newton method. The TCIM possesses superior convergence properties and can be used to analyze distribution networks with high R/X ratios. Moreover, the TCIM can analyze distribution feeders with single-phase and multiphase DER penetration because of the three-phase analysis feature. This work develops the TCIM-based CPF tool to analyze voltage stability of radial and meshed distribution networks with high DER penetration.

A. TCIM Equations

The specified power for PV's real power and PQ's real and reactive power is represented in (1) and (2), where, for each phase, s and k are the nodes of interest [24]

$$(P_k^{\text{spec}})^s = P_{gk}^s - P_{lk}^s \tag{1}$$

$$(Q_k^{\text{spec}})^s = Q_{gk}^s - Q_{lk}^s. \tag{2}$$

The calculated power is for PV's real power and PQ's real and reactive power shown in (3), and (4) was arrived at by expanding the power equation in [23] for a three-phase interpretation

$$(P_k^{\text{cal}})^s = \sum_{i=1}^{N} \sum_{t \in abc} |V_k||V_i|[G_{ki}^{st} \cos(\delta_{ki}^{st}) + B_{ki}^{st} \sin(\delta_{ki}^{st})] \tag{3}$$

$$(Q_k^{\text{cal}})^s = \sum_{i=1}^{N} \sum_{t \in abc} |V_k||V_i|[G_{ki}^{st} \sin(\delta_{ki}^{st}) - B_{ki}^{st} \cos(\delta_{ki}^{st})]. \tag{4}$$

where

N	= number of buses;
r, m	= real, imaginary;
$s, t \in$	$[a, b, c]$;
δ_k^s, δ_k^i	voltage angle of phase s/phase i;
δ_{ki}^{st}	$= \delta_k^s - \delta_i^t$;
$Y_{ki}^{st} = G_{ki}^{st} + jB_{ki}^{st}$	element of a nodal admittance matrix.

The power mismatch is performed for all three phases, which is equal to the power specified taken out of the power calculated, as seen in

$$(\Delta P_k)^s = (P_k^{\text{spec}})^s - (P_k^{\text{cal}})^s \tag{5}$$

$$(\Delta Q_k)^s = (Q_k^{\text{spec}})^s - (Q_k^{\text{cal}})^s. \tag{6}$$

The current mismatch is determined for all PV and PQ buses (excludes slack bus). The form for the current mismatch equation depends on the type of bus, shown in (7)–(9). The current mismatch for PQ buses comes from [23] and [25] and the mismatches for PV buses is from [23] and [24]. All three references share a commonality with slight differences. Paper [25] does not use (7) and (8) since it uses a power mismatch

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for PV buses and current mismatch for PQ nodes, which is a hybrid method of the TCIM.

Current mismatch for PV:

$$(\Delta I_{mk})^s = \frac{(V_{mk})^s \Delta P_k^s}{(V_k^2)^s} \tag{7}$$

$$(\Delta I_{rk})^s = \frac{V_{rk}^s \Delta P_k^s}{(V_k^2)^s} \tag{8}$$

current mismatch for PQ:

$$(\Delta I_{mk})^s = \frac{V_{mk}^s \Delta P_k^s - V_{rk}^s \Delta Q_k^s}{(V_k^2)^s} \tag{9}$$

$$(\Delta I_{rk})^s = \frac{V_{rk}^s \Delta P_k^s + V_{mk}^s \Delta Q_k^s}{(V_k^2)^s} \tag{10}$$

where $(V_k^2)^s = (V_{rk}^s)^2 + (V_{mk}^s)^2$.

The Jacobian, seen in (11), has PQ components that are constant, while the PV components change. The following equations were formulated using [23] and [24]:

$$\text{Jacobian} = J = \begin{bmatrix} Y_{22}^{abc} & \cdots & Y_{2n}^{abc} \\ \vdots & \ddots & \vdots \\ Y_{n2}^{abc} & \cdots & Y_{nn}^{abc} \end{bmatrix} \tag{11}$$

where n is the number of nodes and Y_{jk}^{abc} is a 6×6 matrix formed by the nodes k and j. With different correlating equations depending on whether the node is PV or PQ, we represent PV nodes in Jacobian as follows.

When $j = k$ and k is a PV node, we have

$$Y_{jk}^{abc} = Y_{kk}^{abc} = \begin{bmatrix} M & O \\ N & P \end{bmatrix} \tag{12}$$

where

$$O = \begin{bmatrix} \frac{V_{rk}^a}{(V_k^a)^2} & 0 & 0 \\ 0 & \frac{V_{rk}^b}{(V_k^b)^2} & 0 \\ 0 & 0 & \frac{V_{rk}^c}{(V_k^c)^2} \end{bmatrix} \tag{13}$$

$$P = \begin{bmatrix} \frac{-V_{mrk}^a}{(V_k^a)^2} & 0 & 0 \\ 0 & \frac{-V_{mrk}^b}{(V_k^b)^2} & 0 \\ 0 & 0 & \frac{-V_{mrk}^c}{(V_k^c)^2} \end{bmatrix} \tag{14}$$

$$M = \begin{bmatrix} G_{kk}^{aa} - B_{kk}^{aa}\left(\frac{V_{mrk}^a}{V_{rk}^a}\right) & G_{kk}^{ab} - B_{kk}^{ab}\left(\frac{V_{mrk}^a}{V_{rk}^a}\right) \\ G_{kk}^{ba} - B_{kk}^{ba}\left(\frac{V_{mrk}^b}{V_{rk}^b}\right) & G_{kk}^{bb} - B_{kk}^{bb}\left(\frac{V_{mrk}^b}{V_{rk}^b}\right) \\ G_{kk}^{ca} - B_{kk}^{ca}\left(\frac{V_{mrk}^c}{V_{rk}^c}\right) & G_{kk}^{cb} - B_{kk}^{cb}\left(\frac{V_{mrk}^c}{V_{rk}^c}\right) \end{bmatrix}$$

$$\begin{matrix} G_{kk}^{ac} - B_{kc}^{ac}\left(\frac{V_{mrk}^a}{V_{rk}^a}\right) \\ G_{kk}^{bc} - B_{kk}^{bc}\left(\frac{V_{mrk}^b}{V_{rk}^b}\right) \\ G_{kk}^{cc} - B_{kk}^{cc}\left(\frac{V_{mrk}^c}{V_{rk}^c}\right) \end{matrix} \Bigg] \tag{15}$$

$$N = \begin{bmatrix} -B_{kk}^{aa} - G_{kk}^{aa}\left(\frac{V_{mrk}^a}{V_{rk}^a}\right) & -B_{kk}^{ab} - G_{kk}^{ab}\left(\frac{V_{mrk}^a}{V_{rk}^a}\right) \\ -B_{kk}^{ba} - G_{kk}^{ba}\left(\frac{V_{mrk}^b}{V_{rk}^b}\right) & -B_{kk}^{bb} - G_{kk}^{bb}\left(\frac{V_{mrk}^b}{V_{rk}^b}\right) \\ -B_{kk}^{ca} - G_{kk}^{ca}\left(\frac{V_{mrk}^c}{V_{rk}^c}\right) & -B_{kk}^{cb} - G_{kk}^{cb}\left(\frac{V_{mrk}^c}{V_{rk}^c}\right) \end{bmatrix}$$

$$\begin{matrix} -B_{kc}^{ac} - G_{kk}^{ac}\left(\frac{V_{mrk}^a}{V_{rk}^a}\right) \\ -B_{kk}^{bc} - G_{kk}^{bc}\left(\frac{V_{mrk}^b}{V_{rk}^b}\right) \\ -B_{kk}^{cc} - G_{kk}^{cc}\left(\frac{V_{mrk}^c}{V_{rk}^c}\right) \end{matrix} \Bigg]. \tag{16}$$

When j is not equal to k and k is a PV node, we have

$$Y_{jk}^{abc} = \begin{bmatrix} Q & U \\ R & W \end{bmatrix}. \tag{17}$$

where U and W are zero matrices ($U = W = [0]$) and

$$Q = \begin{bmatrix} G_{jk}^{aa} - B_{jk}^{aa}\left(\frac{V_{mrk}^a}{V_{rk}^a}\right) & G_{jk}^{ab} - B_{jk}^{ab}\left(\frac{V_{mrk}^a}{V_{rk}^a}\right) \\ G_{jk}^{ba} - B_{jk}^{ba}\left(\frac{V_{mrk}^b}{V_{rk}^b}\right) & G_{jk}^{bb} - B_{jk}^{bb}\left(\frac{V_{mrk}^b}{V_{rk}^b}\right) \\ G_{jk}^{ca} - B_{jk}^{ca}\left(\frac{V_{mrk}^c}{V_{rk}^c}\right) & G_{jk}^{cb} - B_{jk}^{cb}\left(\frac{V_{mrk}^c}{V_{rk}^c}\right) \end{bmatrix}$$

$$\begin{matrix} .. \; G_{jk}^{ac} - B_{jc}^{ac}\left(\frac{V_{mrk}^a}{V_{rk}^a}\right) \\ .. \; G_{jk}^{bc} - B_{jk}^{bc}\left(\frac{V_{mrk}^b}{V_{rk}^b}\right) \\ .. \; G_{jk}^{cc} - B_{jk}^{cc}\left(\frac{V_{mrk}^c}{V_{rk}^c}\right) \end{matrix} \Bigg] \tag{18}$$

$$R = \begin{bmatrix} -B_{jk}^{aa} - G_{jk}^{aa}\left(\frac{V_{mrk}^a}{V_{rk}^a}\right) & -B_{jk}^{ab} - G_{jk}^{ab}\left(\frac{V_{mrk}^a}{V_{rk}^a}\right) \\ -B_{jk}^{ba} - G_{jk}^{ba}\left(\frac{V_{mrk}^b}{V_{rk}^b}\right) & -B_{jk}^{bb} - G_{jk}^{bb}\left(\frac{V_{mrk}^b}{V_{rk}^b}\right) \\ -B_{jk}^{ca} - G_{jk}^{ca}\left(\frac{V_{mrk}^c}{V_{rk}^c}\right) & -B_{jk}^{cb} - G_{jk}^{cb}\left(\frac{V_{mrk}^c}{V_{rk}^c}\right) \end{bmatrix}$$

$$\begin{matrix} -B_{jc}^{ac} - G_{jk}^{ac}\left(\frac{V_{mrk}^a}{V_{rk}^a}\right) \\ -B_{jkj}^{bc} - G_{jk}^{bc}\left(\frac{V_{mrk}^b}{V_{rk}^b}\right) \\ -B_{jk}^{cc} - G_{jk}^{cc}\left(\frac{V_{mrk}^c}{V_{rk}^c}\right) \end{matrix} \Bigg]. \tag{19}$$

For all other situations as long as k is not a PV node, we have

$$Y_{jk}^{abc} = \begin{bmatrix} B_{jk}^{aa} & B_{jk}^{ab} & B_{jk}^{ac} & G_{jk}^{aa} & G_{jk}^{ab} & G_{jk}^{ac} \\ B_{jk}^{ba} & B_{jk}^{bb} & B_{jk}^{bc} & G_{jk}^{ba} & G_{jk}^{bb} & G_{jk}^{bc} \\ B_{jk}^{ca} & B_{jk}^{cb} & B_{jk}^{cc} & G_{jk}^{ca} & G_{jk}^{cb} & G_{jk}^{cc} \\ G_{jk}^{aa} & G_{jk}^{ab} & G_{jk}^{ac} & -B_{jk}^{aa} & -B_{jk}^{ab} & -B_{jk}^{ac} \\ G_{jk}^{ba} & G_{jk}^{bb} & G_{jk}^{bc} & -B_{jk}^{ba} & -B_{jk}^{bb} & -B_{jk}^{bc} \\ G_{jk}^{ca} & G_{jk}^{cb} & G_{jk}^{cc} & -B_{jk}^{ca} & -B_{jk}^{cb} & -B_{jk}^{cc} \end{bmatrix}. \tag{20}$$

The change in in real and imaginary voltages is found from (21) as follows:

$$\begin{bmatrix} \Delta I_{m2}^{abc} \\ \Delta I_{r2}^{abc} \\ \vdots \\ \Delta I_{mn}^{abc} \\ \Delta I_{rn}^{abc} \end{bmatrix} = \begin{bmatrix} Y_{22}^{abc} & \cdots & Y_{2n}^{abc} \\ \vdots & \ddots & \vdots \\ Y_{n2}^{abc} & \cdots & Y_{nn}^{abc} \end{bmatrix} \begin{bmatrix} \Delta V_{r2}^{abc} \\ \Delta V_{m2}^{abc} \\ \vdots \\ \Delta V_{rn}^{abc} \\ \Delta V_{mn}^{abc} \end{bmatrix} \tag{21}$$

$$\boldsymbol{\Delta} I = J \Delta V$$

where k is a PV node, n is the number of nodes, and

$$\Delta I_{m2}^{abc} = \begin{bmatrix} \Delta I_{m2}^a \\ \Delta I_{m2}^b \\ \Delta I_{m2}^c \end{bmatrix}.$$

Once ΔV has been found, the real (V_r) and imaginary (V_m) voltages are updated. The update for PQ nodes are as expected and shown in (22) from [24].

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Algorithm 1: TCIM Power Flow Algorithm.

1: Obtain network data and initialize iteration and voltages.
2: Calculate the three-phase power mismatches from (5) and (6)
3: Check if the power mismatches are within tolerance, then stop; else go to step 5.
4: Compute the TCIM Jacobian matrix using (11)
5: Solve for change in real and imaginary voltages from (21)
6: Update real and imaginary voltages from (22)–(24); go to step 2

For PV nodes, the voltage magnitude is held constant, while the angles are updated. The imaginary part of the voltage, thus, needs to be updated. The new real part of the voltage is found using the updated imaginary voltage and the specified voltage magnitude at the PV bus. These equations are seen in (23)–(24).

For PQ nodes, we have

$$(V^s_{rmk})^{(h+1)} = (V^s_{rmk})^{(h)} + (\Delta V^s_{rmk})^{(h)}. \tag{22}$$

For PV nodes, we have

$$(V^s_{mk})^{(h+1)} = (V^s_{mk})^{(h)} + (\Delta V^s_{mk})^{(h)} \tag{23}$$

$$(V^s_{rk})^{(h+1)} = \sqrt{(V^s_k)^2 - \left((V^s_{rmk})^{(h+1)}\right)^2} \tag{24}$$

where s is a single phase, k is a PV node, and V^s_k is the specified voltage for node k at phase s.

B. Algorithm

The three-phase power flow algorithm is presented in Algorithm 1 and explained in Fig. 1.

III. Three-Phase Continuation Power Flow

CPF is used to determine the system loading limit as the load is increased from the base case in small increments. CPF traces the P–V curve using a series of predictor–corrector steps. Nonlinear power flow equations are parameterized in terms of the loading parameter λ to solve the CPF. The parameterized nonlinear power flow equations $f(x, \lambda) = 0$ are solved to obtain V_r (real part of voltages), V_m (imaginary part of voltages), and λ (loading parameter). The CPF discussed in [7] traces the solution curve for any network to determine the voltage collapse point (in power system terms) or the saddle-node bifurcation point (in mathematical terms). Thus, the determined bifurcation point represents the system maximum loading limit. Power injection equations in terms of the loading parameter λ are described as follows:

$$\Delta P_k = (P_g - (1+\lambda)P_l) - P^{\text{cal}}_k \tag{25}$$

$$\Delta Q_k = (Q_g - (1+\lambda)Q_l) - Q^{\text{cal}}_k. \tag{26}$$

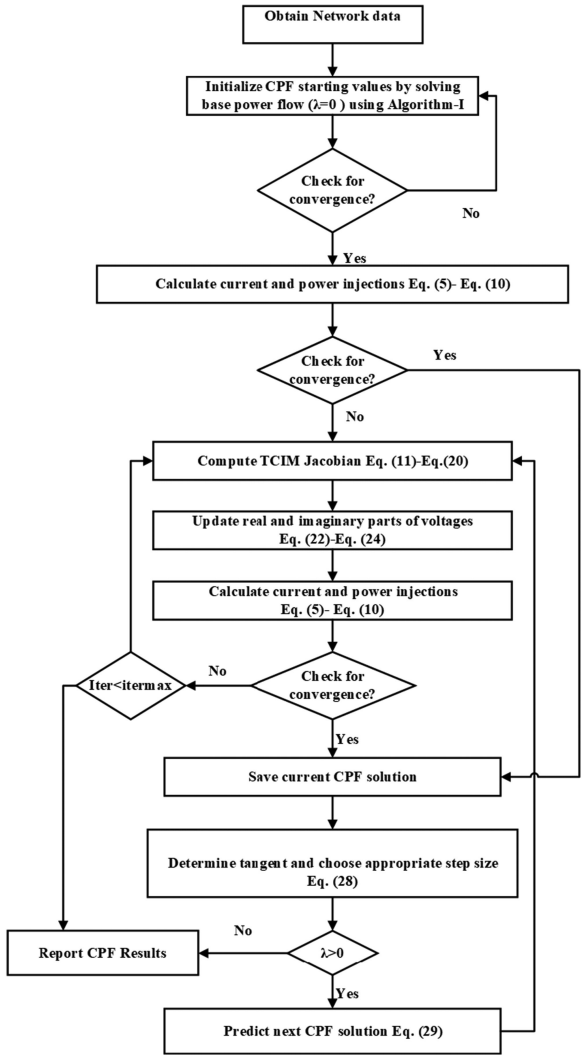


Fig. 1. Three-phase continuation power flow.

A. Predictor Step

The nonlinear power flow equations $f(x) = 0$ are augmented with the loading parameter λ such that the new equations take the following form:

$$f(x, \lambda) = 0. \tag{27}$$

The new state variables for the CPF problem are V_r, V_m, and λ. The predictor determines an approximate solution using the computed tangent and the chosen step length (σ). The tangent $[dV_r,\ \ dV_m,\ \ d\lambda\ ]^T$ is used to predict initial values for the next CPF step. The tangent size is ($6n$ + 1) for a three-phase system, where there are $6n$ voltage variables and one loading parameter (λ) variable. Prediction can be made using tangent and secant predictors. Tangent predictors predict just using the current solution, and the secant predictors use current and previous solutions to make predictions

$$\begin{bmatrix} f_{V_r}(x,\lambda)\ f_{V_m}(x,\lambda)\ f_\lambda(x,\lambda) \\ E_k \end{bmatrix} \begin{bmatrix} dV_m \\ dV_r \\ d\lambda \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \tag{28}$$

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$$\begin{bmatrix} V_m \\ V_r \\ \lambda \end{bmatrix}_{\text{pred}} = \begin{bmatrix} V_m \\ V_r \\ \lambda \end{bmatrix}_{\text{prev}} + \sigma \begin{bmatrix} dV_m \\ dV_r \\ d\lambda \end{bmatrix}. \tag{29}$$

Here, $f_{V_r}(x,\lambda)$, $f_{V_m}(x,\lambda)$, and $f_\lambda(x,\lambda)$ are the derivatives of nonlinear power flow equations with respect to V_r, V_m, and λ, respectively. The row vector E_k is used to augment the Jacobian, and its value depends upon the choice of the continuation parameter.

B. Corrector

The corrector solves the augmented power flow using the estimated solution predicted by the CPF predictor. The augmented power flow problem is given by

$$\begin{pmatrix} f(x,\lambda) \\ \hline x_k - \eta \end{pmatrix} = \begin{bmatrix} 0 \\ \hline 0 \end{bmatrix} \tag{30}$$

where η is an estimated value of the selected continuation parameter x_k. The choice of a continuation parameter would depend on the largest change in the state variables between the last two operating points. Initially, λ is the chosen continuation parameter, but, as the nose point of P–V curve approaches, the voltage of bus showing the most significant change is selected as the continuation parameter.

C. Step Length Control

Step length control helps the CPF algorithm determine the nose point in a short time span. The step length can be constant or adaptive. An adaptive step length gives more flexibility and can avoid convergence issues near the system loading limit. For a faster adaptive step length, the beginning step lengths are larger. The closer the curve approaches the maximum loading point, the small the step sizes become. The step will become smaller and smaller at the nose point as the predictor estimates the next solution. In this work, the adaptive step length is adopted for TPCPF analysis. Initially, the step length is kept constant on the flat part of the P–V curve, and near the nose, the step length is reduced to allow better convergence characteristics and tracing of the P–V curve. Algorithm 2 explains the adaptive step length control for a particular predictor–corrector step.

Algorithm 2: Adaptive Step-Length Control.

1 Initialize predictor-corrector iteration counter (i=0), and specify initial step length ($\sigma^{(i)}, i=0$) to find h^{th} CPF solution.
2 Execute predictor step according to step length at i^{th} predictor-corrector step as follows:

$$x^{(pred)}(h) = x(h-1) + \sigma^{(i)} dx$$

3 Execute the corrector step to compute the hth CPF solution using predicted value as initial value and:
4 **if** *Corrector converges* **then**
5 Save the hth CPF solution and proceed to compute the $h+1^{th}$ CPF solution.
6 **else**
7 **if** $i \leq max_{iter}$ **then**
8 Set, $\sigma^{(i+1)} = \frac{\sigma^{(0)}}{i+1}$

IV. Results and Discussion

In this work, the proposed algorithm is tested on four test systems. The first test system is the IEEE 13-node feeder, which is an unbalanced radial distribution network. The second test system is the 18-bus shipboard system, which is a balanced radial system. The third test system is the 13-bus CIGRE benchmark system, which is a meshed medium-voltage system possessing large amounts of distributed generation. In addition to small meshed and radial distribution networks, the developed tool is also used to demonstrate a large 136-bus radial distribution feeder's voltage stability analysis. Different case studies are considered for validation of the proposed three-phase CPF algorithm, which includes considering the impact of load increase, the impact of the DER model, the impact of unbalance, and the impact of meshed and radial configuration on the system loading capability.

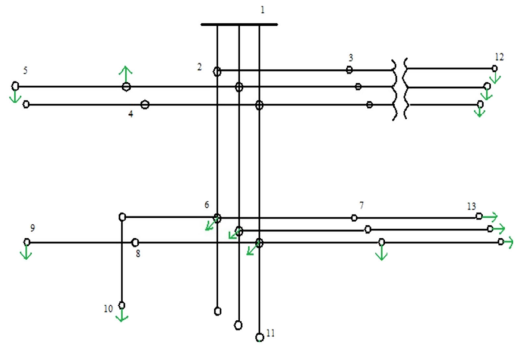


Fig. 2. IEEE 13-bus system.

A. IEEE 13-Bus System

The first test system is the IEEE 13-node feeder, which is an unbalanced radial network with single- and three-phase loads. The network is shown in Fig. 2.

1) Base-Case CPF Results: The P–V curves using the developed tool are obtained, as shown in Fig. 3, for bus 13 in the IEEE 13-bus system. This P–V curve depicts only the upper half, which is the stable portion of the P–V curve. Bus 13 has an unbalanced load connected to the three phases. Thus, the P–V curve for each phase is depicted in Fig. 3 giving each phase's loading capability and voltage stability. It is observed in Fig. 3 that phase c is the most critical phase, which is understandable, as in the base case, phase c is the most loaded phase in the IEEE 13-node feeder.

2) Effect of Adaptive Step Length: The step length in CPF is used to control the speed and convergence characteristics of the continuation process. The step length should be larger when the system is less stressed because convergence is easier and should become lower as the system becomes more stressed and

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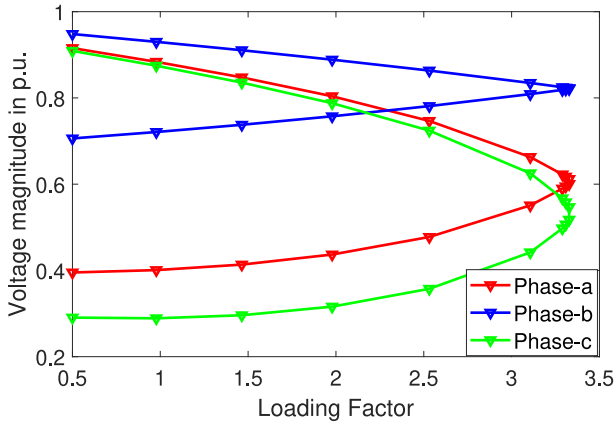


Fig. 3. P–V curves for bus 13.

TABLE I
COMPARISON OF TCM AND CPF RESULTS AT $\lambda = 2.71765$

Case with λ = 2.71765	Voltage Magnitude in p.u.					
	TCIM Results			CPF Results		
Bus	Va	Vb	Vc	Va	Vb	Vc
1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
2	0.8499	0.8994	0.8471	0.8499	0.8994	0.8471
3	0.8388	0.8942	0.8415	0.8388	0.8942	0.8415
4	1.0000	0.8785	0.8471	1.0000	0.8785	0.8471
5	1.0000	0.8716	0.8471	1.0000	0.8716	0.8471
6	0.7436	0.8553	0.7186	0.7436	0.8553	0.7186
7	0.7262	0.8523	0.6949	0.7262	0.8523	0.6949
8	0.7387	1.0000	0.7143	0.7387	1.0000	0.7143
9	1.0000	1.0000	0.7084	1.0000	1.0000	0.7084
10	0.7288	1.0000	1.0000	0.7288	1.0000	1.0000
11	0.7436	0.8553	0.7186	0.7436	0.8553	0.7186
12	0.7091	0.8050	0.7449	0.7091	0.8050	0.7449
13	0.7124	0.8504	0.6851	0.7124	0.8504	0.6851

ultimately reaches the collapse point. In this work, an adaptive step length control is considered, where at the start, a step length of 0.5 is chosen, and as the nose point is approached, the step length is reduced in proportion to the number of iterations taken in the current CPF step. It can be observed that in Fig. 3 initially, a large step length is chosen, and as the nose point is approached, the step length adapts to a smaller value.

3) Validation of CPF Results With the TCIM: The TPCPF tool utilizes the TCIM algorithm for power flow and uses the formulation explained in Section III for the CPF. TCIM results for a particular loading factor are compared with the TPCPF results at that particular loading factor. Table I compares the TPCPF results with TCIM power flow results for the loading factor of $\lambda = 2.71765$. The results obtained validate the performance of the TPCPF power flow solver, which is also based on the TCIM. The specific advantage of TPCPF lies in implementing predictor and corrector steps that ensure convergence till the nose point using a CPF implementation. The predictor–corrector-based implementation enhances the computational efficiency of TPCPF in comparison to the TCIM. Fig. 4 compares the number of iterations taken by the TCIM when the solution is computed by a flat start against TPCPF, which uses predictor–corrector implementation. It is observed that for all the values of λ in Fig. 4, TPCPF takes fewer iterations to converge. It is also observed that at $\lambda = 3.394$, the TCIM fails to converge, whereas

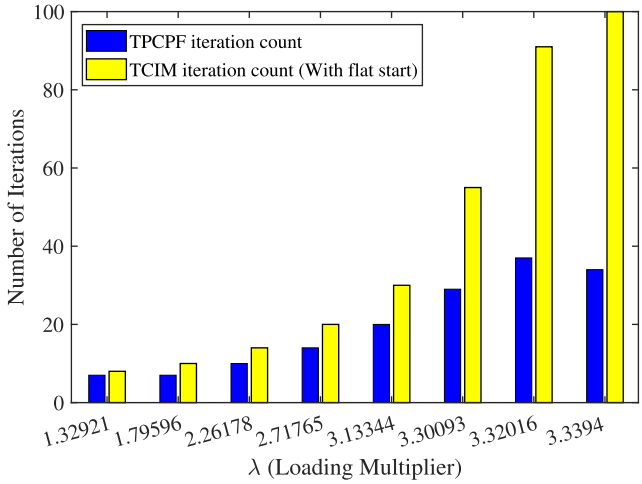


Fig. 4. Computational efficiency comparison: TPCPF versus TCIM (IEEE 13-node feeder).

TABLE II
EFFECT OF LOAD ON THE LOADING CAPABILITY OF THE IEEE 13-BUS SYSTEM

Scenarios	Maximum Loading Factor (λ_{max})	Lowest Voltage at λ_{max}	Lowest Voltage Bus	Lowest Voltage Phase
Case 1	3.3394	0.5403	13	c
Case 2	3.3131	0.5453	13	c
Case 3	3.2527	0.5417	13	c
Case 4	2.6373	0.5190	13	c

TPCPF still converges (maximum number of iterations = 100). This demonstrates that TPCPF possesses superior computational efficiency and convergence characteristics.

4) Effect of Load on the System Loading Capability: To study the impact of load on the maximum loading limit, four different cases are considered:

1) Case 1: base case;
2) Case 2: increase load at node 13 by ten units;
3) Case 3: increase load at node 13 by 50 units;
4) Case 4: increase load at node 13 by 100 units.

Cases 2–4 were evaluated to understand the effect of load increment on the system loading margin without increasing the generation. Table II represents the maximum loading factors for different study scenarios. There is no linear relationship between the load increments and the loading margin change. Also, it can be observed that the increase in load affects the lowest voltage magnitude. Results clearly show that unbalance in the network can greatly impact the voltage stability status.

5) Effect of DER on the Loading Capability of the System: In order to investigate the effect of DERs on the voltage stability of the unbalanced three-phase distribution system, the following case studies were performed:

1) Case 1: base case;
2) Case 2: case 1 with 500-kW DER at bus 2;
3) Case 3: case 1 with 500-kW DER at bus 11;
4) Case 4: case 1 with 500-kW DER at bus 3.

Test results with a PQ-type DER (cases 1–4) are presented in Table III . The results confirm that the installation of DERs

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TABLE III
IMPACT OF DER ON THE LOADING CAPABILITY

Case No.	1	2	3	4
λ_{max}	3.337	3.497	3.768	3.487

TABLE IV
IMPACT OF THE DER MODEL ON THE LOADING MARGIN

DER bus	DER Model	Power rating (kW)	λ_{max}
2	PQ	500	3.497
	PV	500	5.96
11	PQ	500	3.769
	PV	500	5.517
3	PQ	500	3.487
	PV	500	5.192

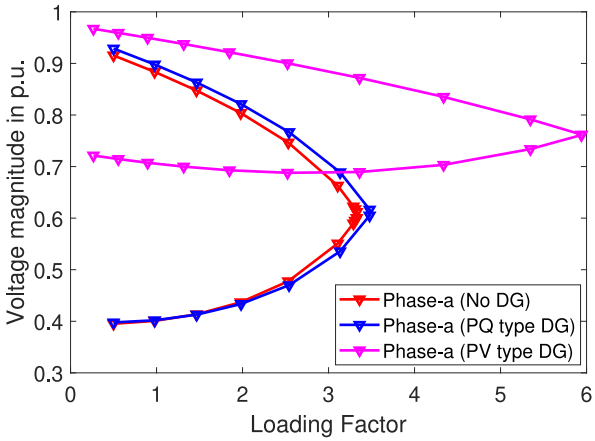


Fig. 5. Voltage profile of bus 13 with DER at bus 2.

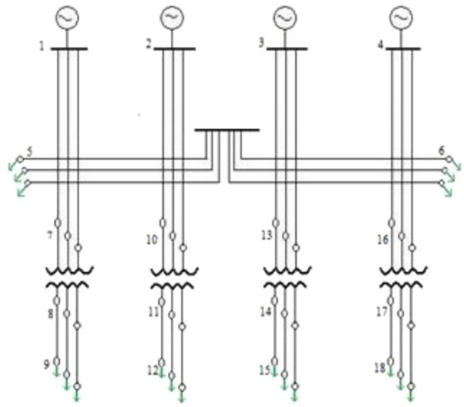


Fig. 6. 18-bus shipboard balanced system.

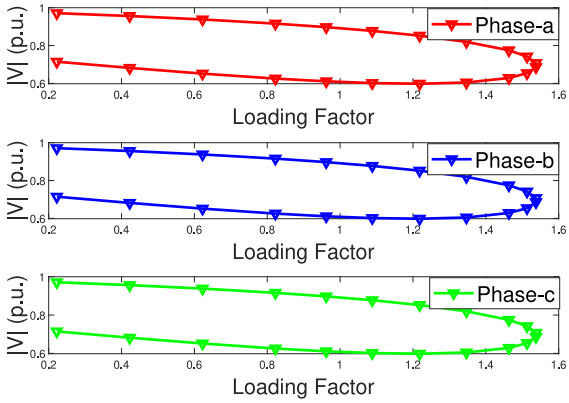


Fig. 7. P–V curve of bus 18.

indeed enhances the loading margin. Location of DERs impacts the overall enhancement in the margin, and DER placement studies considering voltage stability can be performed using the presented tool.

6) Effect of the DER Model on the Loading Capability of the System: The characteristics of DER impact the voltage stability of the distribution network. TPCPF allows PV and PQ representation of DER for CPF analysis. System loading margin is computed for cases 2–4 with the PV-type DER model to demonstrate the impact of the DER model. Table IV compares the system margin with PV and PQ DER models.

7) Effect of DER on the System Voltage Profile: The maximum loading factor for the IEEE 13-bus system based on base-case TPCPF results is 3.337. The DER installation improves the system loading capability, as demonstrated in Table IV. The installation of DER also enhances the system voltage profile. The P–V curves for the weakest bus (bus 13) in the distribution network are shown in Fig. 5, where DER is installed at bus 2. It is observed that the installation of DER significantly improves the voltage profile. Enhancement of voltage profile can reduce system line losses, and such studies can help find the optimal location for DER installation.

In addition to this, the characteristics of DER also impact the voltage profile, and a better voltage profile is observed when a PV-type DER is installed.

B. 18-Bus Shipboard System

The second test system used to demonstrate the effectiveness of TPCPF is the 18-bus shipboard system, which possesses generators, load buses, and transformers, as shown in Fig. 6.

1) Base-Case CPF Results: The P–V curves for bus 18 using the developed tool are shown in Fig. 7. Fig. 7 presents the stable and unstable portion of the P–V curves for all three bus phases. The test system under study is a balanced radial system. Thus, the P–V curve for each phase is depicted in Fig. 7 giving the loading capability, and the voltage stability of each phase was similar. The performance of TPCPF is compared to TCIM in Fig. 8, which confirms the superior computational efficiency and better convergence characteristics of TPCPF.

2) Effect of Load on the Loading Capability of the System: The following cases on the IEEE 18-bus system have been studied:

1) Case 1: base case;
2) Case 2: increase load at node 13 by ten units;
3) Case 3: increase load at node 13 by 50 units;
4) Case 4: increase load at node 13 by 100 units.

Cases 2–4 were evaluated to understand the effect of load increment on the system loading margin without increasing the generation. Table V represents the maximum loading factors for different study scenarios. It can be demonstrated from the

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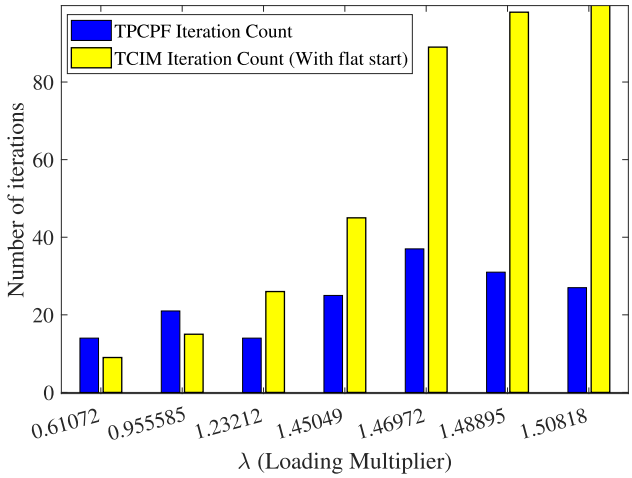


Fig. 8. Computational efficiency comparison: TPCPF versus TCIM (18-bus shipboard system).

TABLE V
EFFECT OF LOAD ON THE LOADING CAPABILITY OF THE 18-BUS SHIPBOARD SYSTEM

Scenarios	Maximum Loading Factor (λ_{max})	Lowest Voltage at λ_{max}	Lowest Voltage Bus	Lowest Voltage Phase
Case 1	1.5045	0.7417	18	a,b,c
Case 2	1.4828	0.7546	18	a,b,c
Case 3	1.4537	0.7594	18	a,b,c
Case 4	1.4217	0.7621	18	a,b,c

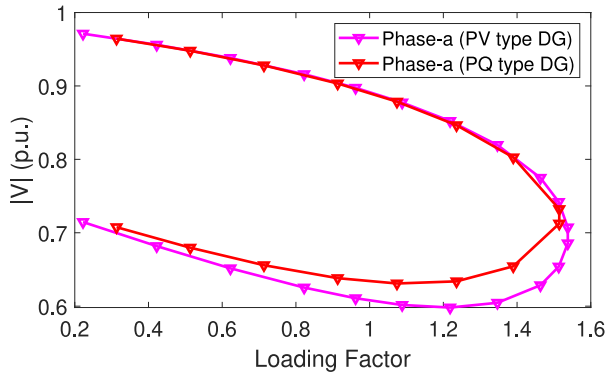


Fig. 9. P–V curve of bus 18—impact of the DER model.

results that the loading capability of the system is reduced with the increment of the load on the system. From the results, it can be clearly seen that the load profile can significantly affect the voltage stability of the system.

3) Impact of the DER Model: The effect of the representation of the generator bus as PQ and PV buses is studied for the 18-bus shipboard system. The P–V curves for bus 18 for both cases of generator bus modeled as PQ and PV buses are shown in Fig. 9. It can be seen that with the PV model, the maximum loading factor of the system is improved, and also, the voltage profile before the nose point is slightly improved. The results are summarized in Table VI.

TABLE VI
IMPACT OF THE DER MODEL ON THE LOADING MARGIN OF THE 18-BUS SHIPBOARD SYSTEM

DER buses	DER Model	Power rating (kW)	λ_{max}
2,3,4	PQ	18.45,18.12,18.18	1.5045
2,3,4	PV	18.45,18.12,18.18	1.545

TABLE VII
IMPACT OF UNBALANCE IN THE 18-BUS SHIPBOARD SYSTEM

Unbalance (%)	λ_{max}	Critical phase	Min. Voltage
0	1.5045	a,b,c	0.7223
2.299	0.9224	b	0.7044
3.67	0.7149	b	0.7019

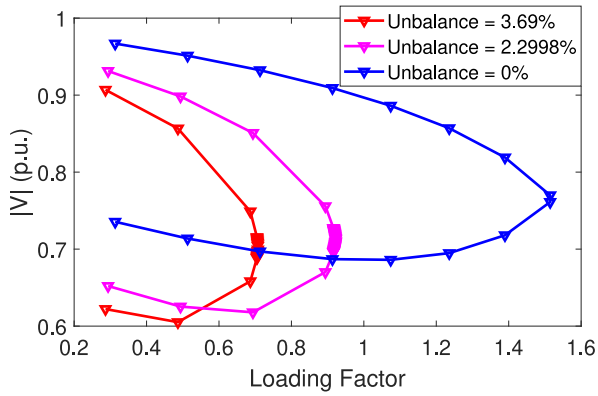


Fig. 10. Impact of unbalance in the (phase-b bus-9) Shipboard Power System (SPS) 18-bus system.

4) Impact of Unbalance on Loading Capability: The amount of unbalance in the distribution system impacts the voltage profile as the loading multiplier is increased from the base case till the maximum loading limit to study the impact of the unbalance percentage unbalance factor is defined as follows:

$$V_{\text{avg}} = \frac{\sum_{i\in[a,b,c]} |V_i|}{3} \tag{31}$$

$$V_{\text{unb}} = \frac{|V_i| - V_{\text{avg}}}{V_{\text{avg}}} \tag{32}$$

$$C_{\text{unb}} = \max(V_{\text{unb}}) \times 100. \tag{33}$$

For the 18-bus system, which is balanced in the base case, unbalance is introduced by changing the load distribution on different phases. It is observed from Table VII and Fig. 10 that for this test system, increase in system unbalance reduces the maximum loading capability because one phase becomes more heavily loaded.

C. CIGRE 13-Bus System

To study a meshed distribution system with the integration of the distributed generation CIGRE benchmark system [26], a medium-voltage system, as shown in Fig. 11, is selected. The generators' input data in this test system are shown in Table VIII.

1) Base-Case CPF Results: Fig. 12 represents the P–V curves for all the individual three phases of the test system for

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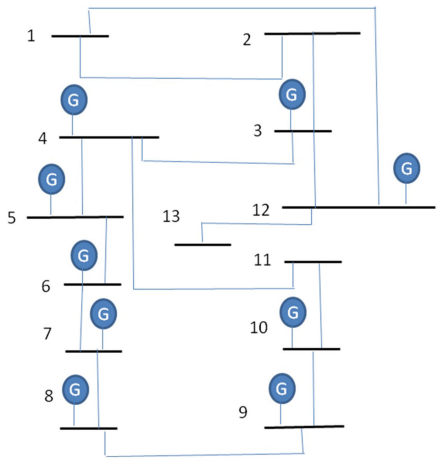


Fig. 11. CIGRE 13-bus benchmark system.

TABLE VIII
GENERATION DATA OF THE CIGRE SYSTEM

Node No.	DER Type	Pmax (kW)
3	Photo voltaic	20
4	Photo voltaic	20
5	Photo voltaic	30
5	Battery	600
5	Fuel Cell in Household	33
6	Photo voltaic	30
7	Wind Turbine	1500
8	Photo voltaic	30
9	Photo voltaic	30
9	CHP Diesel	310
9	CHP Fuel Cell	212
10	Photo voltaic	40
10	Battery	200
10	Fuel Cell in Household	14
11	Photo voltaic	10

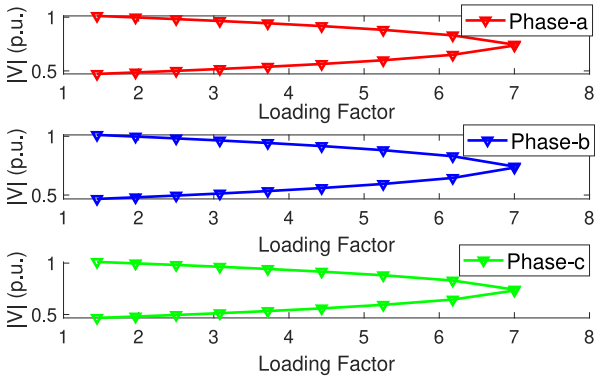


Fig. 12. P–V curve of bus-2 base-case CPF.

bus 2. The system is balanced, so all the three phases of the system have identical P–V curves. The performance of TPCPF is compared to that of the TCIM in Fig. 13, which confirms the superior computational efficiency and better convergence characteristics of TPCPF. As the nose point is approached, the TCIM faces convergence issues (maximum number of iterations $= 100$).

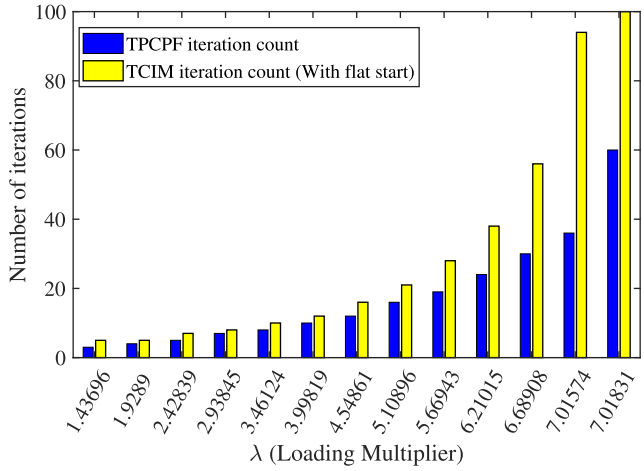


Fig. 13. Computational efficiency comparison: TPCPF versus TCIM (13-bus CIGRE system).

TABLE IX
EFFECT OF LOAD ON THE LOADING CAPABILITY OF THE CIGRE SYSTEM

Scenarios	Maximum Loading Factor (λ_{max})	Lowest Voltage at λ_{max}	Lowest Voltage Bus	Lowest Voltage Phase
Case 1	7.02	0.5635	10	a,b,c
Case 2	6.99	0.5616	10	a,b,c
Case 3	6.87	0.5608	10	a,b,c
Case 4	6.73	0.5534	10	a,b,c

In base-case CPF analysis, all the generators in Fig. 11 are modeled in the PQ (constant power) mode to obtain P–V curves shown in Fig. 12. Similar curves are obtained for all three phases because the CIGRE benchmark is a balanced medium-voltage network.

2) Effect of Load on the System Capability: The following cases on the CIGRE test system have been studied, and it is observed in Table IX that the loading capability is reduced with increment in load:

1) Case 1: base case;
2) Case 2: increase load at node 2 by ten units;
3) Case 3: increase load at node 2 by 50 units;
4) Case 4: increase load at node 2 by 100 units.

3) Impact of the DER Model: The test system is a meshed distribution network. PQ bus representation of the generator in this system provides an improved loading capacity compared to the PV bus model. However, the voltage magnitude for this maximum loading factor (MLF) is very low and can cause voltage collapse in the system. With the PV model, the loading margin is lesser than that in the PQ model case, but the PV model can keep a healthy voltage profile at the bus. It is observed that the lower half of the P–V curve traces in the reverse direction. It can be seen that in this case, with PV-type DERs, unstable portion of the P–V curve of phase a corresponds to the higher voltages. These results agree with the results presented in [14]. Hence, in the meshed test system, the PV-type generators can enhance the overall voltage profile, but the system loading margin can reduce. Higher operating voltages are desirable by distribution system operators for a reliable and efficient system operation.

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TABLE X
IMPACT OF THE DER MODEL ON THE LOADING MARGIN IN THE CIGRE SYSTEM

DER buses	DER Model	λ_{max}
3,4,5,6,7,8,9,11	PV	5.825
3,4,5,6,7,8,9,11	PQ	7.02

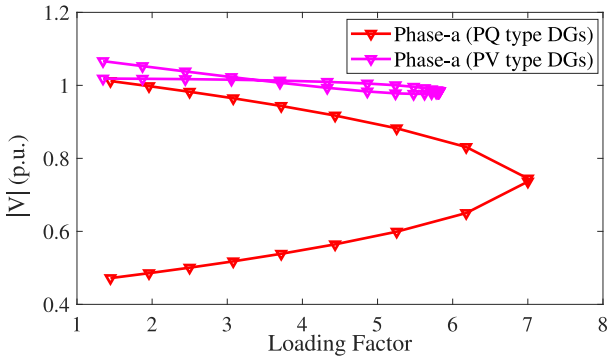


Fig. 14. P–V curve of bus 2—impact of the DER model (CIGRE test system).

TABLE XI
IMPACT OF UNBALANCE IN THE CIGRE SYSTEM

Unbalance (%)	λ_{max}	Critical phase	Min. Voltage
0	7.02	a,b,c	0.5635
0.7957	6.27	b	0.5376
1.2048	5.92	b	0.5429

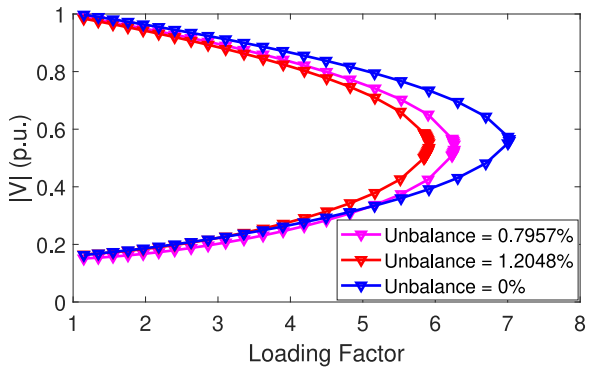


Fig. 15. Impact of unbalance in the (bus-9 phase-b) CIGRE system.

The results are demonstrated in Table X, and P–V curves of bus 2 with PQ- and PV-type DERs are shown in Fig. 14.

4) Impact of Unbalance on Loading Capability: The impact of unbalance on loading capability margin is studied by modeling unbalance in the CIGRE system, which in the base case is a balanced system. The load at bus 9 is distributed such that phase b is loaded more than other phases. It is observed from Table XI and Fig. 15 that an increase in system unbalance results in a reduction of loading capability because one of the phases is loaded more and demonstrates a steeper voltage drop.

D. 136-Bus Radial Distribution System

The 136-bus radial distribution benchmark is used to demonstrate the performance of TPCPF for a large distribution system.

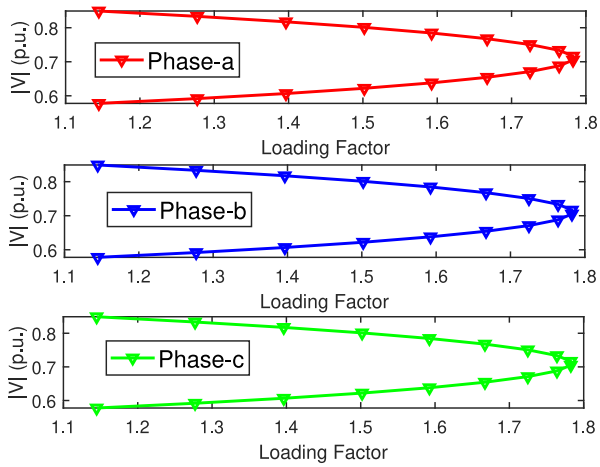


Fig. 16. P–V curves of the bus-2 (136-bus system) base-case CPF.

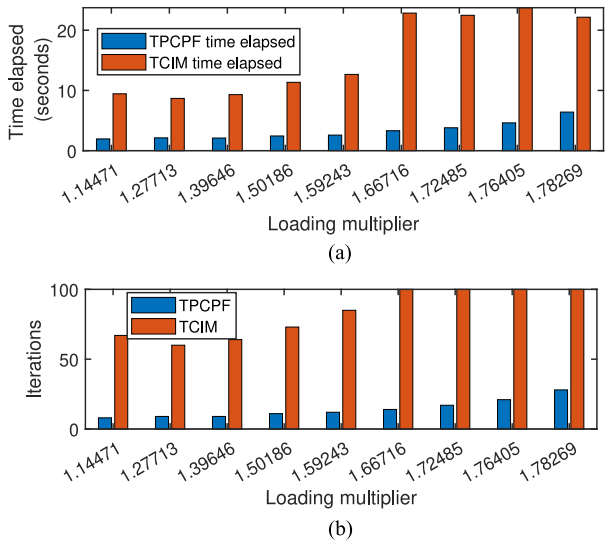


Fig. 17. Computational efficiency comparison (136-bus system). (a) Computational time. (b) Number of iterations.

The considered 136-bus distribution system is a real radial distribution network in Brazil [27]. The network configuration data for the 136-bus radial system are derived from MATPOWER, and it is modeled as a balanced three-phase radial feeder in base-case studies.

1) Base-Case CPF Results: The P–V curves are obtained for bus 16 (phase a) by performing CPF using the developed tool. The P–V curves in Fig. 16 show separate curves for all three phases. Since the network in the base case is balanced, identical curves are obtained for all the phases. The computational performance of TPCPF is compared to that of the TCIM in Fig. 17, which demonstrates the superior performance. The computational efficiency is also compared in terms of computational time taken till convergence at different loading multiplier values.

2) Impact of Unbalance: In order to demonstrate the impact of unbalance on the voltage stability margin, in addition to the

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TABLE XII
IMPACT OF UNBALANCE IN THE 136-BUS REAL DISTRIBUTION SYSTEM

Unbalance (%)	λ_{max}	Critical phase
0	1.798	a,b,c
3.65	1.602	a
7.63	1.44	a

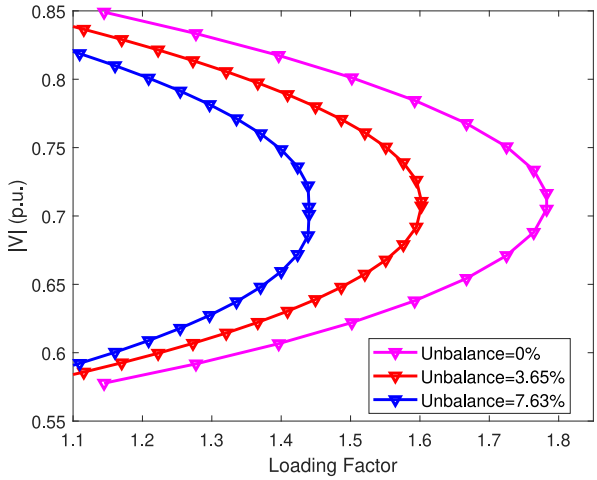


Fig. 18. Impact of unbalance in the (bus-16 phase-a) 136-bus system.

base case, two more case studies are performed with different unbalance levels at base loading. In case 2, at all the buses, the phase-a real power load is increased to 1.1 times the base load, and the real power load of phases b and c is reduced 0.95 times the base loading. An unbalance level of 3.65% is observed in case 2 at base loading. Similarly, in case 3, the real power load at phase a is increased to 1.2 times the base loading level. The phase-c and phase-b real power loads at all the buses are reduced to 0.9 times the base loading in case 3, leading to an unbalanced level of 7.63% at base loading. Table XII and Fig. 18 show the stability limit and P–V curves for the considered case studies. Results show that the maximum loading limit reduces as the unbalance level is increased, and phase a, which is the limiting phase, in this case, results in stability margin reduction.

3) Impact of the DER Model: In order to demonstrate the impact of distributed generation on the voltage stability margin, DERs are modeled in PQ and PV form for CPF analysis. CPF is performed using the developed tool with distributed generators (DGs) installed at buses 2–5. All the DGs inject 250-kW real power and 250-kVAR reactive power at the respective buses. Three different DG configuration cases are considered. In case 1, all DGs are modeled in PQ form. In cases 2 and 3, the DG at bus 2 is modeled in PV form (voltage regulated at 0.93 per unit). In case 2, no reactive power limits are enforced on the DG, whereas in case 3, the reactive power limit of 250 kVAR is enforced on the DG at bus 2. Table XIII presents CPF results and Fig. 19 shows the P–V curves for all the considered cases. Both PQ- and PV-type DGs enhance the voltage profile and the voltage stability margin. It can be observed from Fig. 19 that PV-type DG installation enhances both the voltage profile and the voltage stability margin. When Q-limits are enforced,

TABLE XIII
IMPACT OF THE DER MODEL ON THE LOADING MARGIN IN THE 136-BUS SYSTEM

Case	DER buses	DER Model	λ_{max}
Case-1	2	PQ	1.8235
	3	PQ	
	4	PQ	
	5	PQ	
Case-2	2	PV (Q-limits not enforced)	1.876
	3	PQ	
	4	PQ	
	5	PQ	
Case-3	2	PV (Q-limits enforced)	1.8478
	3	PQ	
	4	PQ	
	5	PQ	

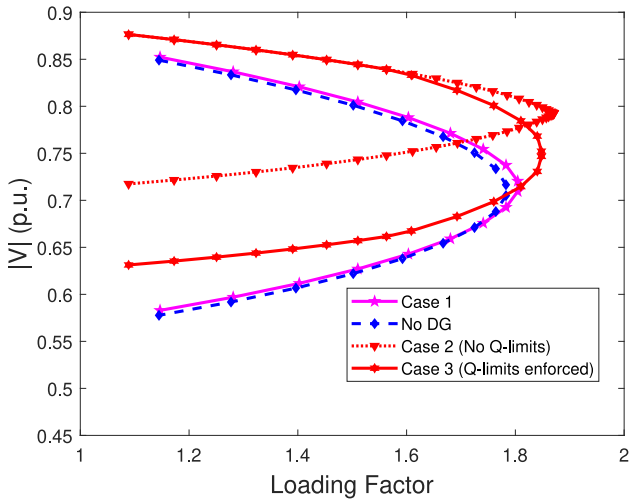


Fig. 19. Impact of the DER model in the (bus-16 phase-a) 136-bus system.

the stability margin is reduced because the DG can no longer supply ample reactive power beyond the Q-limit to keep the voltage constant in the PV mode.

V. CONCLUSION

This work describes development of a three-phase unbalanced CPF algorithm for active distribution systems using the TCIM, adaptive step predictor–corrector, PV–PQ switching, and nose point convergence criteria. The TCIM allows incorporating DERs using a PV (voltage-regulated) or PQ (constant-power) model. This tool can be utilized for the voltage stability assessment of unbalanced distribution feeders with high penetration of DERs. The impacts of load increment on the system for both (balanced and unbalanced) radial and meshed distribution networks have been investigated. PV and PQ models for the DGs have been studied, and their significance is presented with numerical results. Thus, P–V curves obtained using the TPCPF tool can help determine the weakest system bus and provide a guideline for the optimum placement of the distributed generation to improve the voltage stability of the system. The developed TPCPF tool has been validated for multiple test systems, including the IEEE 13-node feeder, the 18-bus balanced shipboard system, the 13-node CIGRE benchmark system, and the 136-bus

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redial distribution feeder for multiple system scenarios. The developed TPCPF can be used effectively for the planning, operation, control, and management of the active distribution systems with high penetration of DERs.

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